Gautham Narayan

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RESEARCH INTERESTS

- Observational Cosmology and Cosmography
- Time-domain Astrophysics, particularly Transient Phenomena
- Wide-field Ultraviolet, Optical and Infrared Surveys
- Multi-messenger Astrophysics & Rapid Follow-up Studies
- Artificial Intelligence, Data Science and Statistics

PROFESSIONAL APPOINTMENTS

Current:	Deputy Director for Astrophysics, NSF-Simons SkAI Institute Sep 2024–present
	Associate Professor, University of Illinois at Urbana-Champaign Aug 2024–present
	Deputy Director, Center for AstroPhysical Surveys, UIUC/NCSA Aug 2020–present
Previous:	Assistant Professor, University of Illinois at Urbana-Champaign Aug 2019–Aug 2024
	Lasker Data Science Fellow, Space Telescope Science Institute Jun 2017–Aug 2019
	Postdoctoral Fellow, National Optical Astronomy Observatory (now NOIRLab) Jul 2013–Jun 2017[1]

EDUCATION

Harvard University	Ph.D. Physics, May 2013 Thesis: "Light Curves of Type Ia Supernovae and Cosmological Constraints from the ESSENCE Survey" Adviser: Prof. Christopher W. Stubbs A.M. Physics, May 2007
Illinois Wesleyan University	B.S. (Hons) Physics, Summa Cum Laude, May 2005 Thesis: "Photometry of Outer-belt Objects" Adviser: Prof. Linda M. French

GRANTS DURING TENURE AT ILLINOIS

- Co-PI, NSF Simons National AI Research Institutes, USD 20,000,000, Sep 2024–present
- Co-PI, NSF Astronomy and Astrophysics Grant (2510994), USD 387,701, Aug 2025–present
- PI, LSST Discovery Alliance Meeting Award, USD 7,500, Aug 2025–present
- PI, NSF Windows on the Universe (2432428), USD 2,363,871, Aug 2024–present
- PI, LSST Discovery Alliance LINCC Incubator, USD 22,500, Aug 2024–present
- PI, LSST Discovery Alliance Workshop, USD 7,500, May 2025–present
- PI, NSF Cyberinfrastructure for Sustained Scientific Innovation (2311355), USD 3,554,724, Aug 2023–present
- PI, NSF CAREER (2239364), USD 717,600, Aug 2023–present
- Co-I, US DoE High Energy Theory & Cosmology, USD 2,901,000, Apr 2023–present
- PI, NSF Astronomy and Astrophysics Grant (2206195), USD 328,618, Aug 2022–present
- PI, HST GO 16764, USD 274,723, Aug 2022–present
- PI, LSST-Wasabi Enabling Science Award, USD 25,000, Aug 2022–present

[1] Formally employed by The University of Arizona CS Dept. from Dec 2014–Apr 2016, but located at NOAO

- PI, LSST Enabling Science Awards for undergraduates at UIUC, USD 10,000, Fall 2021
- PI, NASA ADAP, USD 334,711, 2020–2
- PI, LSST Enabling Science Award, USD 27,000, 2020–2
- Co-I, HST GO 17128 USD 82,030, Oct 2022–present
- Co-I, NCSA Internal Award, "ANTARES at NCSA", USD 88,683, 2020–2
- Co-I on two NASA ADAP proposals, USD 66,835 to UIUC (PIs. D. Jones, A. Rest) 2021–present
- Co-I, grant for developing ANTARES, Heising-Simons Foundation, USD 567,000, 2018–21

AWARDS

- Lincoln Excellence for Assistant Professors, 2023
- Selected for the 1st cohort of the Illinois Academic Leadership and Management Institute
- List of Instructors rated Excellent, Spring 2023
- List of Instructors rated Outstanding, Spring 2020
- 2nd ever recipient of the Barry M. Lasker Data Science Fellowship, STScI, 2017–9
- Best-in-Show, Art of Planetary Science, Lunar and Planetary Laboratory, U. Arizona, 2015

RESEARCH HISTORY AND SELECTED PUBLICATIONS

Below are brief descriptions of my work on key topics, together with a related publication.

Cosmology and the Nature of Dark Energy

- **Spokesperson**, LSST Dark Energy Science Collaboration (DESC, Oct 2025-27), previously Analysis Coordinator (2023-25), Deputy Analysis Coordinator (2021-23) and Supernova Working Group Convener (2019-21).
- Led analysis using ESSENCE and literature SNIa to derive cosmological constraints on dark energy equation of state w; co-authored analysis on PS1 SNIa Foundation photometric SNIa
- Co-developed BayeSN with K. Mandel — probabilistic model to infer distance moduli, light curve, and dust properties from UV+Optical+NIR data of low-z SNIa; currently evolving into model for SNIa SED inference at cosmological distances for *NGRST*, future surveys
- Leading analysis apply BayeSN-SED to combined Foundation, Pan-STARRS & literature samples — will be the largest dataset of confirmed cosmological SNIa

Narayan et al., '16, "Light Curves of 213 Type Ia Supernovae from the ESSENCE Survey", ApJS

Understanding the Physics of Rare and Unusual Transients

- Led or made major contributions to several projects studying unusual SN, including SN 2020oi, SN 2018oh, SN 2009ku, SN 2008ha, as well as open-source tools to model such events
- Combining work on machine learning with interest in rare & unusual transients by developing novel methods for anomaly detection, incorporating gravitational wave, neutrino and high-energy gamma ray signals into alert-brokers.

Narayan et al., '11, "Displaying the Heterogeneity of the SN 2002cx-like Subclass of Type Ia Supernovae with Observations of the Pan-STARRS-1 Discovered SN 2009ku", ApJL

Machine Learning for Time-Domain Discovery

- Lead developer of deep-learning, traditional ML and active learning algorithms for time-series classification on state-of-the-art alert broker system: ANTARES operating on Zwicky Transient Facility (ZTF) alerts as test bed for Legacy Survey of Space and Time (LSST)
- Lead for Extended LSST Astronomical Time-Series Classification Challenge (ELAsTiCC). (v1) remains the largest simulation of of the entire time-domain sky for statistical studies, and the largest Kaggle challenge in Astronomy.

Narayan et al., '18, "Machine-learning-based Brokers for Real-time Classification of the LSST Alert Stream", ApJS Special Issue "Data: Insights and Challenges in a Time of Abundance"

Operations, Calibration, Infrastructure & Optimization of Wide-field Surveys

- PI of SCiMMA Team, developing real-time system to coordinate LIGO, high-energy particle and neutrino, ground- and space-based facilities

- Extensive involvement in transient pipeline development & validation several current and past astrophysical surveys
- Lead analysis to use *Hubble* imaging and large-aperture spectroscopy to establish faint spectrophotometric standards for LSST and future surveys

Narayan et al., '19, "Sub-percent Photometry: Faint DA WD Spectrophotometric Standards for Astrophysical Observatories", ApJS

PROFESSIONAL AFFILIATIONS

I am an active member of several groups and projects, completed and on-going:

The LSST Dark Energy Science Collaboration (DESC)
The SCiMMA Team
The ANTARES Project
The Gravity Wave Collective
The *Kepler* & *TESS* Extra Galactic Survey
The Mosaic z-band Legacy Survey (MzLS)
The LSST Transient & Variable Stars Collaboration (TVS)
The Pan-STARRS PS1 Science Collaboration
The VENUS Collaboration
The [P/E]LAsTiCC Team
The Young Supernova Experiment (YSE)
The DECam Alliance for Transients
The *HST* SIRAH & Dust Survey
The DA White Dwarf Calibration Team
The SN Lightechoes Team
The Foundation Supernova Survey
The ESSENCE Collaboration
The NEXUS Colloboration

Member of the American Astronomical Society (2007–present)

OBSERVING EXPERIENCE

I am an observational cosmologist with extensive experience with different facilities:

- PI, *HST* GO-16764, (34 orbits)
- PI, Gemini N (2020A-Q-115, 2021B-Q-310), Gemini S (2021B-Q-317), programs continue but now led by my students
- Co-PI, Young Supernova Experiment, ∼15% of total time on Pan-STARRS PS1 & PS2, 2020–2027
- PI on two NOIRLab Young Supenova Program on DECam with Long-term Status (LTS) for 2x2 years (2x30 nights)
- Co-I on several major *JWST* ToO programs and *HST* programs with *WFC3*, *ACS* and *STIS* including: GO-12967 (18 orbits), 12999 (8 orbits), 13046 (100 orbits), 13711 (60 orbits), 14216 (100 orbits), 14244 (8 orbits) and 15113 (54 orbits), 17128 (105 orbits)
- PI and Co-I on numerous accepted NOIRLab, Las Cumbres Observatory, ESO and *Swift* observing proposals
- MMT Observatory: 15 nights of Blue Channel spectroscopy on site, 2 nights of remote observing
- Magellan Observatory: 7 nights LDSS3 imaging and long-slit spectroscopy
  1 night of IMACS long-slit spectroscopy
- Gemini Observatory: Analysis of GMOS spectroscopy from several nights of queue observing
- Kitt Peak National Observatory: several nights of imaging on the 4 m with MOSAIC 1.1 & 3
- Cerro-Tololo Inter-American Observatory: several nights with 0.9 m & 4 m with DECam
  Analysis of 197 nights of MOSAIC-II imaging for ESSENCE/SuperMacho
- WIYN Observatory: 3 nights of imaging on the WIYN 3.5 m with ODI
- F. L. Whipple Observatory: several nights of long-slit spectroscopy on the 1.5 m with FAST
  and imaging on the 1.2 m with Keplercam, both on-site and remote
- Las Cumbres Observatory: Analysis of 120 hours of 1 m SINISTRO imaging

MENTORING AND TEACHING

Graduate Students

Athena Engholm (UIUC Astro), Co-adviser, Aug. 2025–present

- Working with Hubble Fellow, Dr. Jason Hinkle, Prof. Decker French, and Narayan on Tidal Disruption Events with Rubin and YSE

Henna Abunemeh (UIUC Astro), Adviser, Aug. 2024–present

- Collaborating with Dr. Daniel Muthukrishna, Dr. Alex Gagliano and Dr. Yunyi Shen to develop encoder methods for SELDON foundation model at SkAI Institute.
- Working with Narayan on YSE spectroscopic sample, and mapping SN diversity, particularly with spectroscopic twin supernovae
- Previous REU student in the UIUC IDEAS program with Narayan in Summer 2023

Padmavathi Venkatraman (UIUC Astro), Co-adviser, Aug. 2024–present

- Co-advising with Prof. Yue Shen (UIUC) on strong lens detection in *JWST* NEXUS Program
- Working with Narayan and Phil Marshall (SLAC) on time-delay cosmology with strongly lensed quasars in Rubin/LSST
- Working on synthetic source injection of lensed quasars into Rubin Data Preview 1 data to measure efficieny

Aadya Agrawal (UIUC Astro), Adviser, Aug. 2023–present

- Leading analysis of lens model consistency paper with SN Hope
- Working with Narayan and Dr. Justin Pierel (STScI) on strongly lensed SNIa
- Lead author on paper comparing lens models on SN Hope [6, accepted], co-author on SN Hope H_0 analysis (Pierel et al. '24)
- Working on ground+space-baased analysis of *HST* SIRAH program supernovae

Tanner Murphy (UIUC Astro), Adviser, Aug. 2023–present, Research Collaborator, Aug. 2020-May 2021

- Working with Narayan on YSE DECAM data, and leading YSE Data Release 2, mentoring undergrads Julia Henricks and Aditya Arunachalam
- Leading analysis of Rubin Data Preview 1 SNe with DECam data
- Previous Masters at SUNY Stonybrook working with Prof. Will Farr on transients
- Advised by Prof. Brian Fields and worked with Narayan to study distribution of SNe within our Galaxy [98]

Haille Perkins (UIUC Astro), Adviser, 2023–present

- Led paper on detectability of KNe without GW triggers with Rubin and *UVEX* [3, accepted]
- Led paper with Prof. Brian Fields on determining lethality of nearby KNe
- DOE SCGSR 2026 at LBNL, Google Summer of Code Program 2025, TARDIS Summer School 2025
- Mentoring Maximillian Riccardi (Bronx Science Academy)

Amanda Wasserman (UIUC Astro), Adviser, 2021–present

- Leading development of SASSAFRAS package to simulate multi-modal transient dataset for SELDON effort at SkAI Institute
- Working on developing RESSPECT active learning and real-time followup for LSST, supervising undergrad Arjun Chainani and Aditya Arunachalam
- Member of LSST DESC & LSST commissioning team, and member of YSE and DECAT teams
- Won DOE SCGSR Award to work on RESSPECT at LBNL in Spring 2024, LSST Wasabi (2022), LSST LINCC Incubator (2024), and Fiddler Innovation Fellowships (2025)

Patrick Aleo (UIUC), Adviser, 2020–2024

- Defended Ph.D., March 2024; initially at US National Security Campus, Kansas City; now data scientist at a private financial corporation
- Winner of Illinois Survey Science Fellowship (2020, 2021) and the Chu Award, 2023
- Worked on anomaly detection for transients within ZTF and YSE [15, LAISS]
- Led 1st data-release of YSE [22], combining data from three telescopes, using AI classification

Alex Gagliano (UIUC), Adviser, 2019–2023

- Defended Ph.D., April 2023, began as NSF IAIFI Fellow between MIT/Harvard, Fall 2023
- Winner of American Statistical Association Astrophysics Interest Group Paper of the Year, 2021
- Winner of the Simons Fellowship CCA (2021), NSF GRF (2020) and Illinois Survey Science Fellowship (2019)
- Worked on correlations between supernovae and their hosts [29, GHOST], SN 2020oi [25], LSST PLAsTiCC, [21, SCOTCH], and other projects.

Daniel Muthukrishna (U. Cambridge), Co-Adviser, 2017–2021

- Scientist at Harvard/CfA, Fall 2025–present
- Staff at MIT, working on *TESS* with Prof. George Ricker, 2021–5
- Research with Narayan on deep learning for transient classification [34, RAPID]
- Previously graduate student at University of Cambridge, co-advised by Mandel and Narayan

Other Mentorship of Graduate Students

- Working closely with Ben Boyd (Cambridge, w/ Prof. Kaisey Mandel), Jiezhong Wu (Northwestern IEMs, w/ Profs Noelle Samia and Dan Apley), and previously with Dr. Yiqi Xie (U. Toronto), Dr. Qinan Wang (MIT) and Dr. Stephen Thorp (Stockholm)
- Served/serving on the prelim/thesis committee for Alejandro Cardenas-Avendano, Scott Perkins, Yiqi Xie (in UIUC Physics), Aidan Berres, Colin Burke, Nicholas Earl, Achintya Krishnan, Margaret Sheperd, Chris Tandoi (Chair), Sunny Tang, Maggie Verrico, David Vizgan, Yujie Wan, Qiaoya Wu (UIUC Astro), Ana Sagues Carracedo (Oskar Klein Center, Stockholm University, Sweden), Willow Fox Fortino (U. Delaware) and Malema Hendrick Ramonyai (University of the Western Cape, S. Africa)
- 2021 summer research advisor for NCSA NRT Summer students Laura Salo (also co-author on ELAsTiCC papers) and Sai Sharan Sundar (together with Eliu Huerta) both from U. Minnesota
- Mentorship of grad students from the U. Arizona Computer Science Dept. on ANTARES, particularly Zhe Wang and Shuo Yang (2015–7)

Undergraduate Students

Aditya Arunachalam (UIUC Astrophysics), Adviser, Nov. 2025-present

- Working with graduate student Tanner Murphey on real-time YSE DECam reductions
- Working with graduate student Amanda Wasserman on characterizing performance of DASH spectroscopic classification algorithm

Julia Henricks (UIUC Astrophysics), Adviser, Nov. 2025-present

- Working with graduate student Tanner Murphey on YSE DECam analysis

Ray Liang (UIUC Astrophysics), Adviser, Nov. 2024-present

- Working with graduate student Aadya Agrawal on fast photometric redshift methods for transients

Tanmay Raswant (UIUC Astrophyiscs + Computer Science), Adviser, Nov. 2024-present

- Working with graduate student Aadya Agrawal on *HST* SIRAH supernovae analaysis

Michael James Lukaszyk (UIUC Astro), Adviser, Oct. 2024-present

- Working with postdoc Dr. Jack O'Brien on modeling type IIP supernovae
- Applying TARDIS to well-observed type IIP SNe from Young Supernovae experiment, building on previous work by J. Vazquez

Gauri Nair (UIUC Physics, Computer Science), Adviser, Oct. 2024-present

- Working with grad student Tanner Murphy on YSE data release 2
- Reducing YSE DECam data with `photpipe` to find young supernovae
- Adapting ZTF BRAAI real/bogus classifier to DECam

Arjun Chainani (UIUC ECE), Adviser, Aug. 2024-present

- Working with grad students Amanda Wasserman and Haille Perkins on RESSPECT framework for LSST Dark Energy Science Collaboration as NCSA SPIN Intern
- Modified RESSPECT to incorporate new ORACLE hierarchical classifier [9]
- Testing RESSPECT as a framework for rapid spectroscopic follow-up of anomalous transients
- With Wasserman, won Fiddler Innovation Fellowship (2025)

Mai Li (UIUC Astro), Adviser, Aug. 2023-May. 2024

- Worked on modeling *HST/STIS* spectroscopy, *HST/WFC3* photometry and ground-based spectroscopy of DA white dwarfs
- Collaborated with graduate students Aidan Berres and Ben Berres (Cambridge) to hierarchically model DA white dwarf sample, testing consistency with previous analysis [8]
- Summer REU at Cornell on gravitational waves

Ved Shah (UIUC CS), Adviser, Jan. 2021–May 2024

- Graduated May 2024, won dept. Wyatt award, now Astronomy Ph.D. at Northwestern, Fall 2024
- Led hierchical real-time classification model for LSST ELAsTiCC [9, ORACLE]
- Led kilonova rate analysis using Bayesian Inference Framework [11, 18]
- Developed M-dwarf flare model with Narayan and postdoc Malanchev; being integrated into PLAsTiCC
- NCSA Summer 2021 Intern; funded by LSST Enabling Science Program (PI Narayan)

Jason Vazquez (UIUC CS+Astro), Adviser, Feb. 2021–May 2024

- Graduated, May 2023, now Physics Ph.D. at UW Milwaukee, Fall 2024
- Finished YSE type IIp sample with Narayan and Kilpatrick
- First-authored paper on SN2019mhm with Kilpatrick and Wong
- Worked on SN2021blg and 2022 YSE supernovae with other survey members
- 2021 Summer REU at Northwestern University with Prof. Wen-fai Fong & C.D. Kilpatrick

Qifeng Cheng (UIUC Astro), Adviser, Feb. 2021–May 2023

- Graduated May 2023, won dept. Wyatt award, now PhD in Astronomy at Duke University in Fall 2023
- Developed new dwarf nova and changing-look AGN model with Narayan and postdocs Soraisam & Malanchev
- Northwestern REU with Patrick Sheehan, Summer 2021, funded by LSST Enabling Science Program (PI Narayan)
- Earned honors with Narayan for work in ASTR 210 by developing viz. of analema for exoplanetary systems

Sammy Sharief (UIUC CS+Astro), Adviser, Feb. 2021–May 2023

- Began Masters in Computer Science at U. Montreal in Fall 2023, Ph.D. at Universite Paris Cite, Fall 2025
- Graduated May 2023, completed senior thesis on work at TAURUS program with Stella Offner (UIUC adviser, Narayan)
- Third author on YSE DR1 with Narayan and Patrick Aleo, and implemented YSE DECam pipeline [22]
- Adapted method to retrieve ZTF forced photometry from IPAC & modeling SNe
- REU on Multi-messenger Astrophysics at RIT, Summer 2021 and at TAURUS program, UT Austin, Summer 2022

Filip Matasic (UIUC CS+Astro), Adviser, Sep. 2021–May 2022

- Graduated May 2022, completed a senior honors thesis with Narayan, now working in InfoSec
- Worked with Aleo on YSE 1st data release [22]
- Analyzed properties of YSE and ZTF observations to generate simulations for ML models

Andrew Engel (UIUC Physics), Adviser, 2019–present

- Beginning Ph.D. at The Ohio State University as Dean's Fellow, Fall 2024
- Graduated May 2020 – went on to position as Sr. data scientist at PNNL
- Working on machine learning for photometric redshifts from galaxies – awarded NASA ADAP for photo-z work, and continuing to work with Narayan & Eileen Beyer
- First author on multi-survey photometric redshift paper [12, Mantis Shrimp]
- Algorithm developed being used in both ANTARES and YSE

Daniel Alcantara (Bard College), Research Collaborator, 2016–9

- Intern with R. Street at Las Cumbres Observatory working on microlensing detection
- Worked with Narayan to dramatically improve performance of prototype classifier
- Published Alcantara, Bachelet, Narayan and Street, 2019
- Algorithm being used with MARS broker to find microlensing candidates with ZTF

Tayeb Zaidi (Macalester College), Honors Thesis Adviser, 2016–7

- Now in PhD Program for Biophysics at Northwestern University, working on MRI devices
- Worked with Narayan on ANTARES as summer REU student at NOAO in 2015
- Continued work on time-series classification for Senior Honors (earned April 2017)
- Published Narayan, Zaidi, Soraisam et al., 2018, adapted for LSST PLAsTiCC

Co-mentorship of Other Undergraduate Students

- Holly Wingren (UIUC Astronomy, '24) - jointly with Prof. Brian Fields and Alex Gagliano. REU at Fermilab & UIUC Statistics, won Layla Ryan Memorial Award from UIUC Astronomy, starting Masters in Education
- Athish Thiruvengadam (UIUC CS+Astronomy, '24) - jointly with Prof. Eliu Huerta
- Kunal Bhatia (UIUC Astronomy, '21) - jointly with Dr. Monika Soraisam. Now pursuing Masters in Physics at Uni. Heidelberg, Astronomisches Rechen-Institut in the Gravitational Lensing Group

Previous REU Students

- Henna Abunemeh (UIUC IDEAS, 2023) graduated May 2024 from UI Chicago, and is beginning her Ph.D. at UIUC, Fall 2024, having completed work on a DOE SULI REU at Argonne National Lab with Dr. Lindsey Bleem
- Marcus Lee (NOAO, 2014) was the first indigenous (Tohono O'odham) student to complete REU program at NOAO

- Linoy Kotler (STScI, 2018) worked on wavelet-based classification of Foundation photometric SNIa sample and is now at Rice University

Postdoctoral Scientists Mentored

Jason Hinkle (UIUC, SkAI), Aug 2025–present

- Hubble Fellow (2025) based at NSF-Simons SkAI Institute and CIERA, working on Tidal Disruption Events with Prof. Decker French, Prof. Adam Miller (NU) and graduate student, Athena Engholm

Jack O'Brien (UIUC), Aug 2024–present

- Co-developed SELDON Foundation AI model for supernova property inference, anomaly detection and forecasting as part of NSF-Simons SkAI Institute [2, SELDON]
- Working with Narayan and grad student Amanda Wasserman on modeling supernovae photometry and spectroscopy to understand physical properties and progenitor systems

Ayan Mitra (UIUC), Nov 2023–present

- Illinois Survey Science Postdoctoral Fellowship in 2023, also serves as a Dark Energy Science Collaboration (DESC) Pipeline Scientist
- Working with Narayan on SNe Ia cosmology analysis for the Young Supernova Experiment and DESC
- Leading analysis of photometrically classified SNe Ia sample with photometric redshifts
- Lead author on first fully-photometric SN Ia cosmology pipeline for LSST [4]

Konstantin Malanchev (UIUC), 2020–2023

- Now LINCC Research Scientist at Carnegie-Mellon University, Summer 2023–present
- Worked with Narayan and Patrick Aleo on anomaly detection, YSE DR1, developing feature extraction & ELAsTiCC
- Developing large cross-matched, cross-calibrated photometric database for LSST
- Worked with undergrad Qifeng Cheng on changing-look AGN model

Deep Chatterjee (UIUC), 2020–2022

- Second recipient of Illinois Survey Science Postdoctoral Fellowship
- Worked with Narayan on ANTARES on kilonovae detection and integration with SCiMMA
- Worked on using deep-learning for rapid approximation of neutron star EoS and identification of electromagnetic counterpart (El-Cid)
- Advised undergrad Ved Shah Gautam along with Narayan on estimating KN rates
- Now Alerts and Real-time Inference Lead at MIT LIGO Lab

Monika Soraisam (UIUC), 2019–2021

- First recipient of Illinois Survey Science Postdoctoral Fellowship
- Working with Narayan on ANTARES (since 2016), and extreme time-domain phenomena
- PI of NCSA CDDR grant to deploy ANTARES broker system on Radiant (Co-I Narayan)
- Worked with undergrads Qifeng Cheng on dwarf nova model, and Kunal Bhatia on ZTF transients
- Now Research Staff at Gemini Observatory

Instructor for:

ASTR 496, Foundations of Data Science in Astronomy, Fall 2025
ASTR 414, Astronomical Techniques, Fall 2024
ASTR 350, The Big Bang, Black Holes, and the End of the Universe, Fall 2023
ASTR 310, Computing in Astronomy, UIUC, Fall 2021, Spring 2022, Fall 2022
ASTR 210, Introduction to Astrophysics, UIUC, Fall 2020
ASTR 596, Fundamentals of Data Science, UIUC, Spring 2020, Spring 2023
ZTF Summer School, Pasadena, Aug. 2018
LSST Data Science Fellowship Program, Session 5, Baltimore, Jan. 2018
LSST Data Science Fellowship Program, Session 3, Tucson, Apr. 2017
NOAO Teen Astronomy Cafe, "How Stars Die", Tucson, Nov. 2017
NOAO Big Data Workshop for Tucson High School Students, Tucson, Jan. 2017
Python Workshop for NOAO/NSO REU Students, Tucson, Summer 2014 & 2015

I've additionally served as a Teaching Assistant at Harvard, as Teaching Assistant, Lab Assistant and Tutor at Illinois Wesleyan, and as guest lecturer for Astro 102 (Instructors: C. Salyk and K. Garmany) at the Tohono O'odham Community College.

INVITED COLLOQUIA/SEMINARS/CONFERENCES, 2019–PRESENT

Invited Panelist, AI Panel, NOIRLab Data-Rich Astronomy Conference, Tucson, AZ, Apr. 2026
Invited Speaker, Transients in Middle Earth, Chirstchurch, NZ, Feb. 2026
University of Chicago, KICP, Chicago, Colloquium, Jan. 2026
Panelist, American Astronomical Society, AIxAstronomy Panel, Jan. 2026
Kavli Institute of Particle Astrophysics and Cosmology, Menlo Park, Dec. 2025
Plenary Speaker, ML4Jets Conferences, California Institute of Technology, Pasadena, Aug. 2025
Plenary Speaker, Cosmic Lighthouses Conference, University of Cambridge, Cambridge, UK, July 2025
Invited Speaker, European Astronomical Society Meeting LSST Session, Cork, Ireland, Jun. 2025
Invited Speaker, Foundation Models for Astrophysics (FM4Astro), Flatiron Institute, New York, May 2025
Invited Panelist, International Post-Exascale (InPEx) Conference, Yokohama, Japan, Apr. 2025
Invited Speaker, National Academy of Sciences Committee on Astronomy and Astrophysics, Apr. 2025
Northwestern University, CIERA, Evanston, Colloquium, Feb. 2025
University of Wisconsin, Milwaukee, Astrophysics Seminar, Dec. 2024
Invited Speaker, Fast Machine Learning for Science, Purdue University, Oct. 2024
Invited Speaker, Gravitational Wave Physics and Astronomy Workshop, Birmingham, UK, May 2024
Invited Speaker, Hot Wiring the Transient Universe VII, University of Toronto, Canada, May 2024
Invited Speaker, Time-Domain Needles in Rubin's Haystacks, Harvard University, Apr. 2024
Invited Speaker, LSST Information and Statistics Science Collaboration, Harvard University, Apr. 2024
Special Seminar, DARK, Copenhagen, Denmark, Feb. 2024
Invited Speaker, What Was That? ESO in the Era of LSST, Garching, Germany, Jan. 2024
Invited Speaker, Cosmic streams in the era of Rubin, Puerto Varras, Chile, Dec. 2023
University of Pennsylvania, Dept. of Physics and Astronomy, Colloquium, Oct. 2023
Invited Speaker, Windows on the Universe, Tucson, Oct. 2023
Indiana University, Dept. of Astronomy, Colloquium, Sep. 2023
Invited Speaker, LSST Project and Community Workshop, Aug. 2023
Invited Speaker, SuperNova EXplosions: Theory and Observations, Technion, Israel, Aug. 2023
Invited Speaker, MARS Lecture Series, Pacific Northwest National Labs, May 2023
Invited Speaker, Multi-wavelength follow-up of Fast Radio Bursts, Toronto, Apr. 2023
Session Chair, Supporting Computational Science with Rubin LSST, Virtual, Mar. 2023
Panelist, The Future of Astrophysical Data Infrastructure, New York, Feb. 2023
Speaker, Panelist, Session Chair, AAS 241, Seattle, Jan. 2023
Dark Energy Science Collaboration, Plenary Talk, University of Chicago, July 2022
Speaker, Bayesian Deep Learning for Cosmology and Time Domain Astrophysics, Paris, Jun 2022
Pennsylvania State University, LSST Workshop, May. 2022
LSST European Broker Coordination Seminar, Apr. 2022
SCiMMA Public Telecon, "A Data Lake for Multi-messenger Astrophysics", Oct. 2021
Pennsylvania State University, Dept. of Astronomy and Astrophysics Colloquium, Mar. 2022
Invited Speaker, Cherenkov Telescope Array Project, Jun 2021
Invited Speaker, AAS 238, Meeting-in a-meeting (MiM) on Machine Learning, Jun 2021
LSST Photometric Calibration Working Group Workshop, May 2021
LSST Broker Workshop, Apr 2021
DESI Time-domain Meeting, Apr 2021
LSST DESC Virtual Meeting - July 2020
SciMMA Virtual Meeting, May 2020
Plenary, Kentucky Area Astronomy Annual Meeting - Louisville, KY, Mar 2020
LSST DESC Annual Meeting - Tucson, AZ, Jan 2020
Kavli Visitor, University of Cambridge, Institute of Astronomy, Dec 2019
Speaker, Session Chair, LSST TVS and SMWLV workshop - Newark, DE, Oct 2019
Speaker, SNIa Cosmology Analysis Meeting - KICP Chicago, IL, Oct 2019

SERVICE & PUBLIC OUTREACH WORK

Service to Scientific Collaborations

Spokesperson, LSST Dark Energy Science Collaboration, Oct 2025–
Interim Rubin Science Collaboration Coordinator, Rubin Observatory, Jul 2024–Jan 2025
Analysis Coordinator, LSST Dark Energy Science Collaboration, Jul 2023–Oct. 2025
Deputy Analysis Coordinator, LSST Dark Energy Science Collaboration, 2021–2023
Executive Committee, Young Supernova Experiment, 2020–present
Member, SN Science Investigation Team, Nancy Grace Roman Space Telescope, 2020–present
Organizer, LSST DESC Hack Week at UIUC, Summer 2022
Co-convener, LSST DESC Supernova Working Group, 2019–2021

Service to the UIUC Astronomy Department

Deputy Director, UIUC/NCSA Center for Astrophysical Surveys, Spring 2021–present
Member, Faculty Search Committee, UIUC Astronomy, 2020, 2021, 2023, 2025 & 2026
Senator, UIUC Faculty Senate, Fall 2022–2024
UIUC AURA Member Representative, 2021–present
Colloquium Chair, UIUC Astronomy, AY 2020, Spring 2023, Fall 2025
Reader, Graduate Admissions Committee, UIUC Astronomy, 2023, 2024
Representative, UIUC Astronomy X Data Science Program, 2022–present
Member, UIUC Astronomy Curriculum Committee, AY 2021–25
Member, UIUC Astronomy Computing Committee, AY 2022–24
Member, UIUC Astronomy EDI Committee, 2020–2023, 2024–25
Member, UIUC Astronomy Development & Donor Relations Committee, AY 2024–25
Primary, UIUC Astronomy Website & Communications / Social Media Committee, AY 2023–25

Meetings and Talk Series

SOC, LSST@Europe 8, Budapest, Hungary, Sep 2026
SOC, European Astronomical Society Annual Meeting, Special Session 27, Lausanne, Switzerland, July 2026
SOC, IAU Symposium 406: The Future Landscape of Astrophysical Transients, Turku, Finland, May 2026
Chair, Rubin Alerts Hackathon 2026, SkAI Institute, Chicago, IL, Apr 2026
SOC, TDA Bench 2026, SkAI Institute, Chicago, IL, Mar 2026
SOC, OpenSkAI Conference, Sep 2025
Chair, LSST Dark Energy Science Collaboration Meeting, UIUC, Urbana, IL, July 2025
SOC, The Era of Binary Supermassive Black Holes, Aspen Center for Physics, Aspen, CO, Feb 2025
Co-Chair, DES-DESC Splinter Meeting at AAS 245, National Harbor, MD, Jan 2025
SOC, Unveiling the Dynamic Universe: Cosmic Streams in the Era of Rubin, Puerto Varas, Chile, Dec 2023
SOC & LOC, The Transient and Variable Universe, UIUC, July 2023
Chair, Boom! A Workshop on LSST Transients, UIUC, July 2022
SOC, COSMO-2021 meeting, Virtual, Aug 2021
SOC, Managing Follow-up Observations in the Era of ZTF and LSST, NSF's NOIRLab, Sep 2019
Chair, Enabling Multi Messenger Astrophysics in the Big Data Era, STScI, Apr, 2019
SOC, Deep Learning for Multimessenger Astrophysics: Real-time Discovery at Scale, Oct 2018
LOC, Building the Infrastructure for Time-Domain Alert Science in the LSST Era, May 2017
Organizer, NOAO FLASH Talk Series, 2015–2017
Organizer, NOAO Coffee Hour Series, 2014–5

Service to the Community

Panelist, NASA Review Panel, Jan-Jun 2022 & Jan-Jun 2024
Panelist, NSF Review Panel, Mar 2020
Editorial Board, Astronomy and Computing, to begin Fall 2023
Reviewer for the AAS Journals, MNRAS, ongoing

Public Outreach

Kaler Lecture, Starkel Planetarium, Apr 2026
Invited Public Talk, "Supernovae Across the Southern Sky with Rubin Observatory", Christchurch, NZ, Feb 2026
Organizer, Astronomy on Tap, Aspen "Fly-by" Event, Feb 2025
Invited Talk and Stargazing, Dark Sky Series, Middle Fork River Forest Preserve, July 2024
Invited Talk, St. Louis Astronomical Society, Washington University, St. Louis, May 2024
Illinois Solar Eclipse, Apr 2024
Invited Talk, Stargazing and S'mores, Allerton Park & Retreat Center, Sep 2023
Inivted Talk, UIAS Open Night, May 2023
Invited Talk, Illinois Dark Skies Star Party, Sangamon Astronomical Society, Oct 2022
Stargazing, Family Night Campout, Allerton Park & Retreat Center, Oct 2022
Invited Talk, University of Illinois Astronomical Society, Sep 2022
Kaler Lecture, Starkel Planetarium - Champaign, IL, Oct 2020
Speaker, Astronomy on Tap - Urbana-Champaign, "The Myth and Mythology of the Planets", Feb 2020
Organizer, Astronomy on Tap - Urbana-Champaign, Nov 2019–Aug 2022
Speaker, Astronomy on Tap - Urbana-Champaign, "Making a Movie of the Night Sky", Sep 2019
Organizer, Astronomy on Tap - Tucson/Space Drafts, 2015-2017
Speaker, Space Telescope Public Lecture Series, Chasing Supernovae with Kepler, Sep 2018
Guest, Three Body Problems Podcast, Bringing Data Science Into Astronomy, Sep 2018
Scientist, TED-Ed Original Videos (Pt. 1) (Pt. 2)
Speaker, 365 Days of Astronomy Podcast (Pt. 1) (Pt. 2)
Speaker, Youth for Astronomy and Engineering, Nov 2018
Speaker, NerdNite Baltimore, Mar 2018
Panelist, Tucson Comic Con and TUSCon, Nov 2015 and 2016
"Robots in Space" and "The Physics of Space Battles"
Speaker, Astronomy on Tap - Tucson with the Tucson Symphony Orchestra, Oct 2016
"A Trip through Gustav Holst's Planets"
Speaker, Astronomy on Tap - Tucson, Jan 2015
"If You Only Knew The Power of The Dark Side"
Speaker, Green Valley Astronomy Club, Sahuarita, AZ, May 2016
Volunteer, Science Night, Elvira Elementary School, Tucson, AZ, Mar 2015 and Mar 2017
Volunteer, Astronomy Night, Arizona Sonoran Desert Museum, July 2015
Volunteer, Kitt Peak National Observatory Open Night for the Tohono O'odham Nation, May 2015
Volunteer, Tucson Festival of Books, Mar 2015

I've led public stargazing at the Museum of Science in Boston (2011–2), the Table Mountain star party, WA (2006) and throughout my time as an undergraduate at Illinois Wesleyan's Mark Evans Observatory (2001–5).

REFERENCES

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LIST OF PUBLICATIONS

h-index: 45, 10,786 citations. (Scopus/Google Scholar)
Publications are listed with 1st author or major contributor first.

Primary Publications

[1] **Attaining Spectral Energy Distributions With Sub-Percent Uncertainties: All-Sky DA White Dwarf Spectrophotometric Standard Stars For Large Telescopes And Surveys**. A. Saha, E. W. Olszewski, B. M. Boyd, T. Matheson, T. Axelrod, **G. Narayan**, A. Calamida, J. B. Holberg, I. Hubeny, R. C. Bohlin, S. Deustua, A. Rest, J. Claver, S. Points, C. W. Stubbs, E. Sabbi, and J. W. Mackenty. *arXiv e-prints*, Mar. 2026. arXiv:2603.10185.

[2] **SELDON: Supernova Explosions Learned by Deep ODE Networks**. J. Wu, J. O'Brien, J. Li, M. S. Krafczyk, V. G. Shah, A. R. Wasserman, D. W. Apley, **G. Narayan**, and N. I. Samia. *arXiv e-prints*, Mar. 2026. arXiv:2603.04392.

[3] **Searching for Neutron Star Mergers in the Absence of Gravitational Waves with Optical Afterglow Emission**. H. M. L. Perkins, **G. Narayan**, B. D. Fields, V. G. Shah, and G. Schroeder. *arXiv e-prints*, Feb. 2026. arXiv:2602.13471.

[4] **A Fully Photometric Approach to Type Ia Supernova Cosmology in the LSST Era: Host Galaxy Redshifts and Supernova Classification**. A. Mitra, R. Kessler, R. C. Chen, A. Gagliano, M. Grayling, S. More, **G. Narayan**, H. Qu, S. Raghunathan, A. I. Malz, M. Lochner, and The LSST Dark Energy Science Collaboration. *arXiv e-prints*, Dec. 2025. arXiv:2512.06319.

[5] **Characterization of type Ibn SNe**. D. Farias, C. Gall, V. A. Villar, K. Auchettl, K. M. de Soto, A. Gagliano, W. B. Hoogendam, **G. Narayan**, A. Sedgewick, S. K. Yadavalli, Y. Zenati, C. R. Angus, K. W. Davis, J. Hjorth, W. V. Jacobson-Galán, D. O. Jones, C. D. Kilpatrick, M. J. Bustamante Rosell, D. A. Coulter, G. Dimitriadis, R. J. Foley, A. Gangopadhyay, H. Gao, M. E. Huber, L. Izzo, J. L. Johnson, A. L. Piro, A. Rest, C. Rojas-Bravo, M. R. Siebert, K. Taggart, and S. Tinyanont. *arXiv e-prints*, Nov. 2025. arXiv:2511.12362.

[6] **Testing Lens Models of PLCK G165.7+67.0 Using Lensed SN H0pe**. A. Agrawal, J. D. R. Pierel, **G. Narayan**, B. L. Frye, J. M. Diego, N. Garuda, M. Grayling, A. M. Koekemoer, K. S. Mandel, M. Pascale, D. Vizgan, and R. A. Windhorst. *arXiv e-prints*, Oct. 2025. arXiv:2510.07637.

[7] **Neural Post-Einsteinian Test of General Relativity with the Third Gravitational-Wave Transient Catalog**. Y. Xie, **G. Narayan**, and N. Yunes. *arXiv e-prints*, Oct. 2025. arXiv:2510.02515.

[8] **DAmodel: Hierarchical Bayesian Modelling of DA White Dwarfs for Spectrophotometric Calibration**. B. M. Boyd, **G. Narayan**, K. S. Mandel, M. Grayling, A. Saha, T. Axelrod, T. Matheson, E. W. Olszewski, A. Calamida, A. Do, R. C. Bohlin, J. B. Holberg, I. Hubeny, S. Deustua, A. Rest, C. W. Stubbs, A. Berres, M. Li, J. W. Mackenty, and E. Sabbi. *Mon. Not. R. Astron. Soc.*, Apr. 2025.

[9] **The Fastest Path to Discovering the Second Electromagnetic Counterpart to a Gravitational Wave Event**. V. G. Shah, R. J. Foley, and **G. Narayan**. *Publ. Astron. Soc. Pac.*, Feb. 2025. 137(2):024101.

[10] **Faint White Dwarf Flux Standards: Data and Models**. R. C. Bohlin, S. Deustua, **G. Narayan**, A. Saha, A. Calamida, K. D. Gordon, J. B. Holberg, I. Hubeny, T. Matheson, and A. Rest. *Astron. J.*, Jan. 2025. 169(1):40.

[11] **ORACLE: A Real-time, Hierarchical, Deep Learning Photometric Classifier for the LSST**. V. G. Shah, A. Gagliano, K. Malanchev, **G. Narayan**, A. I. Malz, and the LSST Dark Energy Science Collaboration. *Astrophys. J.*, Dec. 2025. 995(1):4.

[12] **Mantis Shrimp: Exploring Photometric Band Utilization in Computer Vision Networks for Photometric Redshift Estimation**. A. W. Engel, N. Byler, A. Tsou, **G. Narayan**, E. Bonilla, and I. Smith. *Astrophys. J.*, Oct. 2025. 991(2):124.

[13] **SN 2021foa: The "Flip-flop" Type IIn/Ibn Supernova**. D. Farias, C. Gall, **G. Narayan**, S. Rest, V. A. Villar, C. R. Angus, K. Auchettl, K. W. Davis, R. J. Foley, A. Gagliano, J. Hjorth, L. Izzo, C. D. Kilpatrick, H. M. L. Perkins, E. Ramirez-Ruiz, C. L. Ransome, A. Sarangi, R. Yarza, D. A. Coulter, D. O. Jones, N. Khetan, A. Rest, M. R. Siebert, J. J. Swift, K. Taggart, S. Tinyanont, P. Wrubel, T. J. L. de Boer, K. E. Clever, A. Dhara, H. Gao, and C. C. Lin. *Astrophys. J.*, Dec. 2024. 977(2):152.

[14] **Blast: a Web Application for Characterizing the Host Galaxies of Astrophysical Transients**. D. O. Jones, P. McGill, T. A. Manning, A. Gagliano, B. Wang, D. A. Coulter, R. J. Foley, **G. Narayan**, V. A. Villar, L. Braff, A. W. Engel, D. Farias, Z. Lai, K. Loertscher, J. Kutcka, S. Thorp, and J. Vazquez. *arXiv e-prints*, Oct. 2024. arXiv:2410.17322.

[15] **Anomaly Detection and Approximate Similarity Searches of Transients in Real-time Data Streams**. P. D. Aleo, A. W. Engel, **G. Narayan**, C. R. Angus, K. Malanchev, K. Auchettl, V. F. Baldassare, A. Berres, T. J. L. de Boer, B. M. Boyd, K. C. Chambers, K. W. Davis, N. Esquivel, D. Farias, R. J. Foley, A. Gagliano, C. Gall, H. Gao, S. Gomez, M. Grayling, D. O. Jones, C. C. Lin, E. A. Magnier, K. S. Mandel, T. Matheson, S. I. Raimundo, V. G. Shah, M. D. Soraisam, K. M. de Soto, S. Vicencio, V. A. Villar, and R. J. Wainscoat. *Astrophys. J.*, Oct. 2024. 974(2):172.

[16] **Neural post-Einsteinian framework for efficient theory-agnostic tests of general relativity with gravitational waves**. Y. Xie, D. Chatterjee, **G. Narayan**, and N. Yunes. *Phys. Rev. D.*, Jul. 2024. 110(2):024036.

[17] **Preliminary Report on Mantis Shrimp: a Multi-Survey Computer Vision Photometric Redshift Model**. A. Engel, **G. Narayan**, and N. Byler. *arXiv e-prints*, Feb. 2024. arXiv:2402.03535.

[18] **Predictions for electromagnetic counterparts to Neutron Star mergers discovered during LIGO-Virgo-KAGRA observing runs 4 and 5**. V. G. Shah, **G. Narayan**, H. M. L. Perkins, R. J. Foley, D. Chatterjee, B. Cousins, and P. Macias. *Mon. Not. R. Astron. Soc.*, Feb. 2024. 528(2):pp. 1109–1124.

[19] **Relative Intrinsic Scatter in Hierarchical Type Ia Supernova Sibling Analyses: Application to SNe 2021hpr, 1997bq, and 2008fv in NGC 3147**. S. M. Ward, S. Thorp, K. S. Mandel, S. Dhawan, D. O. Jones, K. Taggart, R. J. Foley, **G. Narayan**, K. C. Chambers, D. A. Coulter, K. W. Davis, T. de Boer, K. de Soto, N. Earl, A. Gagliano, H. Gao, J. Hjorth, M. E. Huber, L. Izzo, D. Langeroodi, E. A. Magnier, P. McGill, A. Rest, C. Rojas-Bravo, R. Wojtak, and Young Supernova Experiment. *Astrophys. J.*, Oct. 2023. 956(2):111.

[20] **Results of the Photometric LSST Astronomical Time-series Classification Challenge (PLAsTiCC)**. R. Hložek, A. I. Malz, K. A. Ponder, M. Dai, **G. Narayan**, E. E. O. Ishida, J. Allam, T., A. Bahmanyar, X. Bi, R. Biswas, K. Boone, S. Chen, N. Du, A. Erdem, L. Galbany, A. Garreta, S. W. Jha, D. O. Jones, R. Kessler, M. Lin, J. Liu, M. Lochner, A. A. Mahabal, K. S. Mandel, P. Margolis, J. R. Martínez-Galarza, J. D. McEwen, D. Muthukrishna, Y. Nakatsuka, T. Noumi, T. Oya, H. V. Peiris, C. M. Peters, J. F. Puget, C. N. Setzer, Siddhartha, S. Stefanov, T. Xie, L. Yan, K. H. Yeh, and W. Zuo. *Astrophys. J. Suppl. Ser.*, Aug. 2023. 267(2):25.

[21] **The simulated catalogue of optical transients and correlated hosts (SCOTCH)**. M. Lokken, A. Gagliano, **G. Narayan**, R. Hložek, R. Kessler, J. F. Crenshaw, L. Salo, C. S. Alves, D. Chatterjee, M. Vincenzi, A. I. Malz, and LSST Dark Energy Science Collaboration. *Mon. Not. R. Astron. Soc.*, Apr. 2023. 520(2):pp. 2887–2912.

[22] **The Young Supernova Experiment Data Release 1 (YSE DR1): Light Curves and Photometric Classification of 1975 Supernovae**. P. D. Aleo, K. Malanchev, S. Sharief, D. O. Jones, **G. Narayan**, R. J. Foley, V. A. Villar, C. R. Angus, V. F. Baldassare, M. J. Bustamante-Rosell, D. Chatterjee, C. Cold, D. A. Coulter, K. W. Davis, S. Dhawan, M. R. Drout, A. Engel, K. D. French, A. Gagliano, C. Gall, J. Hjorth, M. E. Huber, W. V. Jacobson-Galán, C. D. Kilpatrick, D. Langeroodi, P. Macias, K. S. Mandel, R. Margutti, F. Matasić, P. McGill, J. D. R. Pierel, E. Ramirez-Ruiz, C. L. Ransome, C. Rojas-Bravo, M. R. Siebert, K. W. Smith, K. M. de Soto, M. C. Stroh, S. Tinyanont, K. Taggart, S. M. Ward, R. Wojtak, K. Auchettl, P. K. Blanchard, T. J. L. de Boer, B. M. Boyd, C. M. Carroll, K. C. Chambers, L. DeMarchi, G. Dimitriadis, S. A. Dodd, N. Earl, D. Farias, H. Gao, S. Gomez, M. Grayling, C. Grillo, E. E. Hayes, T. Hung, L. Izzo, N. Khetan, A. N. Kolborg, J. A. P. Law-Smith, N. LeBaron, C. C. Lin, Y. Luo, E. A. Magnier, D. Matthews, B. Mockler, A. J. G. O'Grady, Y. C. Pan, C. A. Politsch, S. I. Raimundo, A. Rest, R. Ridden-Harper, A. Sarangi, S. L. Schrøder, S. J. Smartt, G. Terreran, S. Thorp, J. Vazquez, R. J. Wainscoat, Q. Wang, A. R. Wasserman, S. K. Yadavalli, R. Yarza, Y. Zenati, and Young Supernova Experiment. *Astrophys. J. Suppl. Ser.*, May 2023. 266(1):9.

[23] **A hierarchical Bayesian SED model for Type Ia supernovae in the optical to near-infrared**. K. S. Mandel, S. Thorp, **G. Narayan**, A. S. Friedman, and A. Avelino. *Mon. Not. R. Astron. Soc.*, Mar. 2022. 510(3):pp. 3939–3966.

[24] **El-CID: a filter for gravitational-wave electromagnetic counterpart identification**. D. Chatterjee, **G. Narayan**, P. D. Aleo, K. Malanchev, and D. Muthukrishna. *Mon. Not. R. Astron. Soc.*, Jan. 2022. 509(1):pp. 914–930.

[25] **An Early-time Optical and Ultraviolet Excess in the Type-Ic SN 2020oi**. A. Gagliano, L. Izzo, C. D. Kilpatrick, B. Mockler, W. V. Jacobson-Galán, G. Terreran, G. Dimitriadis, Y. Zenati, K. Auchettl, M. R. Drout, **G. Narayan**, R. J. Foley, R. Margutti, A. Rest, D. O. Jones, C. Aganze, P. D. Aleo, A. J. Burgasser, D. A. Coulter, R. Gerasimov, C. Gall, J. Hjorth, C.-C. Hsu, E. A. Magnier, K. S. Mandel, A. L. Piro, C. Rojas-Bravo, M. R. Siebert, H. Stacey, M. C. Stroh, J. J. Swift, K. Taggart, S. Tinyanont, and S. Tinyanont. *Astrophys. J.*, Jan. 2022. 924(2):55.

[26] **The ANTARES Astronomical Time-domain Event Broker**. T. Matheson, C. Stubens, N. Wolf, C.-H. Lee, **G. Narayan**, A. Saha, A. Scott, M. Soraisam, A. S. Bolton, B. Hauger, D. R. Silva, J. Kececioglu, C. Scheidegger, R. Snodgrass, P. D. Aleo, E. Evans-Jacquez, N. Singh, Z. Wang, S. Yang, and Z. Zhao. *Astronomical. J*, Mar. 2021. 161(3):107.

[27] **SN 2018agk: A Prototypical Type Ia Supernova with a Smooth Power-law Rise in Kepler (K2)**. Q. Wang, A. Rest, Y. Zenati, R. Ridden-Harper, G. Dimitriadis, **G. Narayan**, V. A. Villar, M. R. Magee, R. J. Foley, E. J. Shaya, P. Garnavich, L. Wang, L. Hu, A. Bódi, P. Armstrong, K. Auchettl, T. Barclay, G. Barentsen, Z. Bognár, J. Brimacombe, J. Bulger, J. Burke, P. Challis, K. Chambers, D. A. Coulter, G. Csörnyei, B. Cseh, M. Deckers, J. L. Dotson, L. Galbany, S. González-Gaitán, M. Gromadzki, M. Gully-Santiago, O. Hanyecz, C. Hedges, D. Hiramatsu, G. Hosseinzadeh, D. A. Howell, S. B. Howell, M. E. Huber, S. W. Jha, D. O. Jones, R. Könyves-Tóth, C. Kalup, C. D. Kilpatrick, L. Kriskovics, W. Li, T. B. Lowe, S. Margheim, C. McCully, A. Mitra, J. A. Muñoz, M. Nicholl, J. Nordin, A. Pál, Y.-C. Pan, A. L. Piro, S. Rest, J. Rino-Silvestre, C. Rojas-Bravo, K. Sárneczky, M. R. Siebert, S. J. Smartt, K. Smith, Á. Sódor, M. D. Stritzinger, R. Szabó, R. Szakáts, B. E. Tucker, J. Vinkó, X. Wang, J. C. Wheeler, D. R. Young, A. Zenteno, K. Zhang, and G. Zsidi. *Astrophys. J.*, Dec. 2021. 923(2):167.

[28] **Testing the consistency of dust laws in SN Ia host galaxies: a BAYESN examination of Foundation DR1**. S. Thorp, K. S. Mandel, D. O. Jones, S. M. Ward, and **G. Narayan**. *Mon. Not. R. Astron. Soc.*, Dec. 2021. 508(3):pp. 4310–4331.

[29] **GHOST: Using Only Host Galaxy Information to Accurately Associate and Distinguish Supernovae**. A. Gagliano, **G. Narayan**, A. Engel, M. Carrasco Kind, and LSST Dark Energy Science Collaboration. *Astrophys. J.*, Feb. 2021. 908(2):170.

[30] **The Young Supernova Experiment: Survey Goals, Overview, and Operations**. D. O. Jones, R. J. Foley, **G. Narayan**, J. Hjorth, M. E. Huber, P. D. Aleo, K. D. Alexander, C. R. Angus, K. Auchettl, V. F. Baldassare, S. H. Bruun, K. C. Chambers, D. Chatterjee, D. L. Coppejans, D. A. Coulter, L. DeMarchi, G. Dimitriadis, M. R. Drout, A. Engel, K. D. French, A. Gagliano, C. Gall, T. Hung, L. Izzo, W. V. Jacobson-Galán, C. D. Kilpatrick, H. Korhonen, R. Margutti, S. I. Raimundo, E. Ramirez-Ruiz, A. Rest, C. Rojas-Bravo, M. R. Siebert, S. J. Smartt, K. W. Smith, G. Terreran, Q. Wang, R. Wojtak, A. Agnello, Z. Ansari, N. Arendse, A. Baldeschi, P. K. Blanchard, D. Brethauer, J. S. Bright, J. S. Brown, T. J. L. de Boer, S. A. Dodd, J. R. Fairlamb, C. Grillo, A. Hajela, C. Hede, A. N. Kolborg, J. A. P. Law-Smith, C. C. Lin, E. A. Magnier, K. Malanchev, D. Matthews, B. Mockler, D. Muthukrishna, Y. C. Pan, H. Pfister, D. K. Ramanah, S. Rest, A. Sarangi, S. L. Schrøder, C. Stauffer, M. C. Stroh, K. L. Taggart, S. Tinyanont, R. J. Wainscoat, and Young Supernova Experiment. *Astrophys. J.*, Feb. 2021. 908(2):143.

[31] **Models and Simulations for the Photometric LSST Astronomical Time Series Classification Challenge (PLAsTiCC)**. R. Kessler, **G. Narayan**, A. Avelino, E. Bachelet, R. Biswas, P. J. Brown, D. F. Chernoff, A. J. Connolly, M. Dai, S. Daniel, R. Di Stefano, M. R. Drout, L. Galbany, S. González-Gaitán, M. L. Graham, R. Hložek, E. E. O. Ishida, J. Guillochon, S. W. Jha, D. O. Jones, K. S. Mand el, D. Muthukrishna, A. O'Grady, C. M. Peters, J. R. Pierel, K. A. Ponder, A. Prša, S. Rodney, V. A. Villar, LSST Dark Energy Science Collaboration, and Transient and Variable Stars Science Collaboration. *Publ. Astron. Soc. Pac.*, Sep 2019. 131(1003):p. 094501.

[32] **A machine learning classifier for microlensing in wide-field surveys**. D. Godines, E. Bachelet, **G. Narayan**, and R. A. Street. *Astronomy and Computing*, Jul 2019. 28:100298.

[33] **Subpercent Photometry: Faint DA White Dwarf Spectrophotometric Standards for Astrophysical Observatories**. **G. Narayan**, T. Matheson, A. Saha, T. Axelrod, A. Calamida, E. Olszewski, J. Claver, K. S. Mandel, R. C. Bohlin, and J. B. Holberg. *Astrophys. J. Suppl. Ser.*, Apr 2019. 241(2):20.

[34] **RAPID: Early Classification of Explosive Transients Using Deep Learning**. D. Muthukrishna, **G. Narayan**, K. S. Mandel, R. Biswas, and R. Hložek. *Publ. Astron. Soc. Pac.*, Nov. 2019. 131(1005):p. 118002.

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